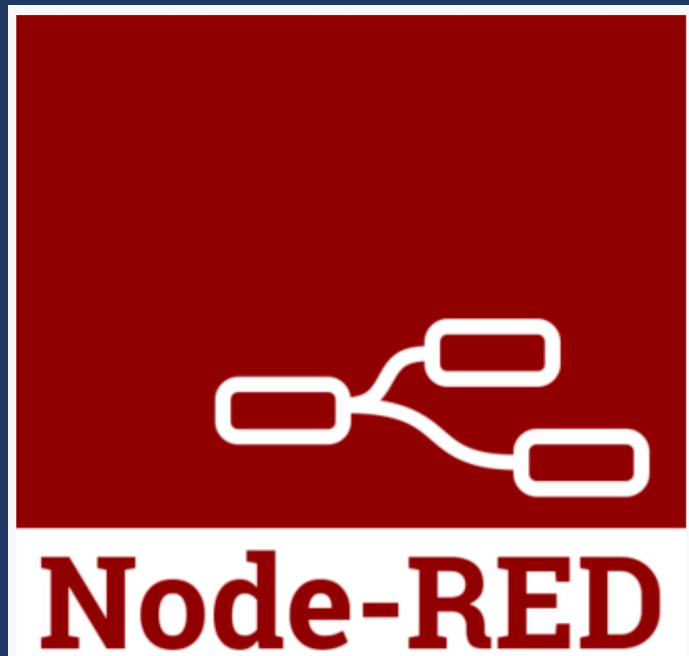




Node-RED: The Basics

Safyzan Salim



V1 JAN 2020 : V2 FEB 2021

Download Node-RED & Install

→ INSTALL Node.js

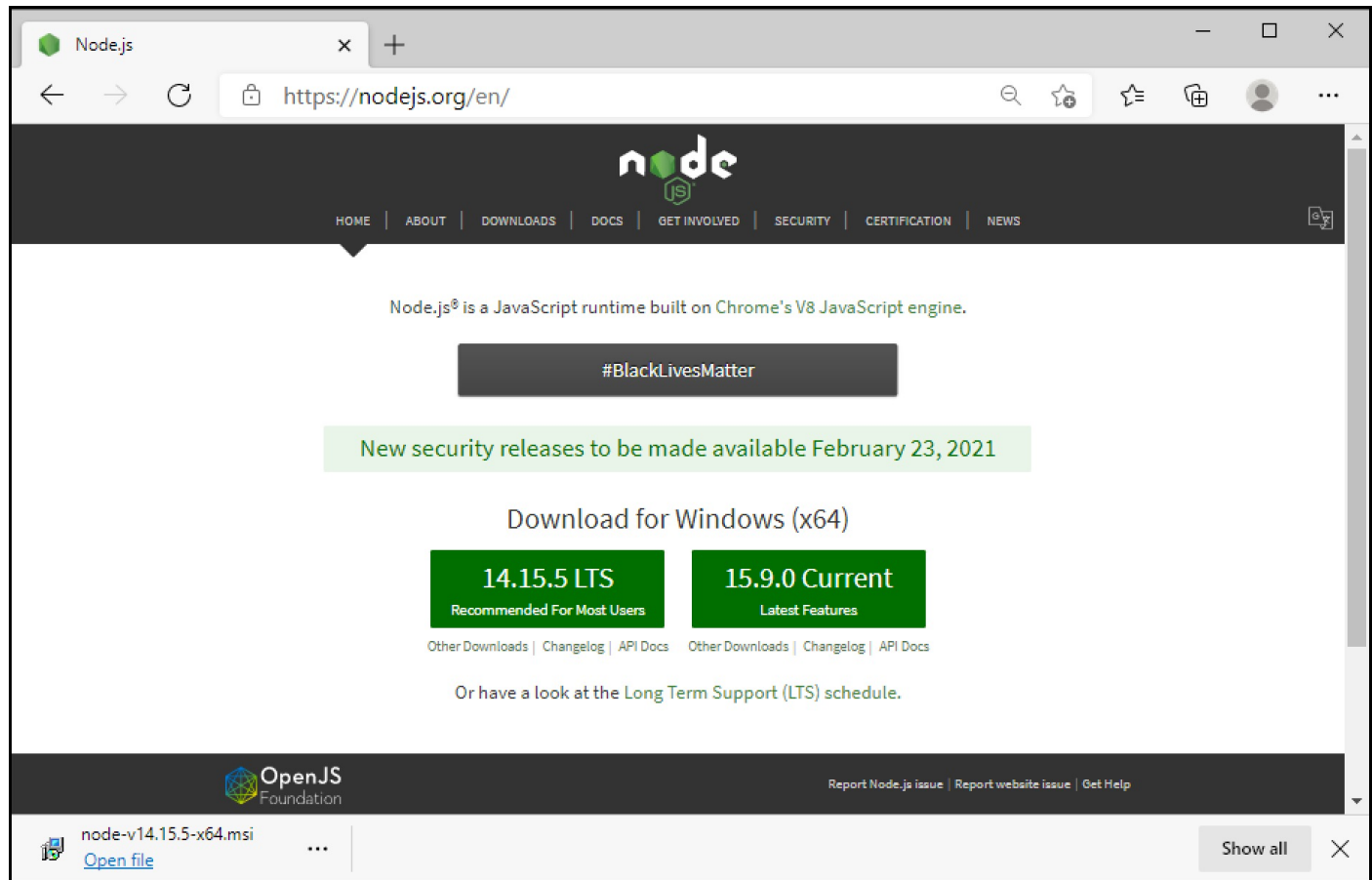
>> Download the latest Long Term Support (LTS) version of Node.js from the official [Node.js](https://nodejs.org/en/) home page. It will offer you the best version for your system.

>> Run the downloaded MSI file.

>> Installing Node.js requires local administrator rights; if you are not a local administrator, you will be prompted for an administrator password on install.

>> Accept the defaults when installing.

>> After installation completes, close any open command prompts and re-open to ensure new environment variables are picked up.



Download Node-RED & Install

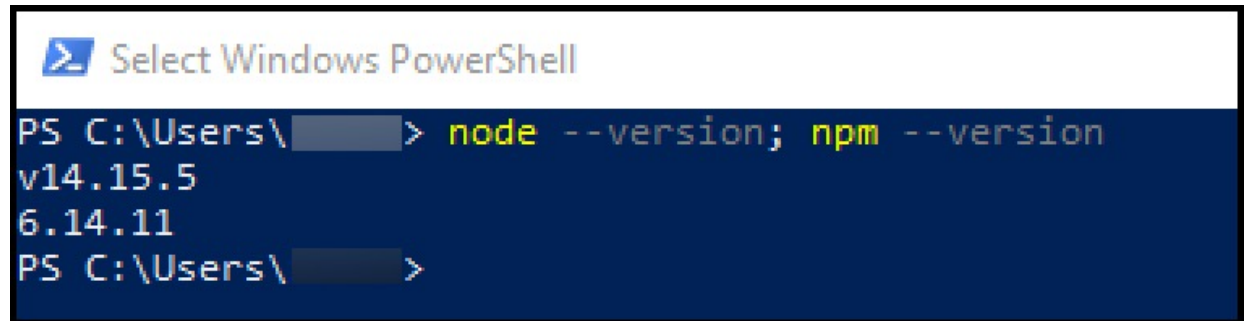
>> Once installed, open a command prompt and run the following command to ensure Node.js and npm are installed correctly.

→ Using Powershell & macOS terminal:

node --version; npm --version

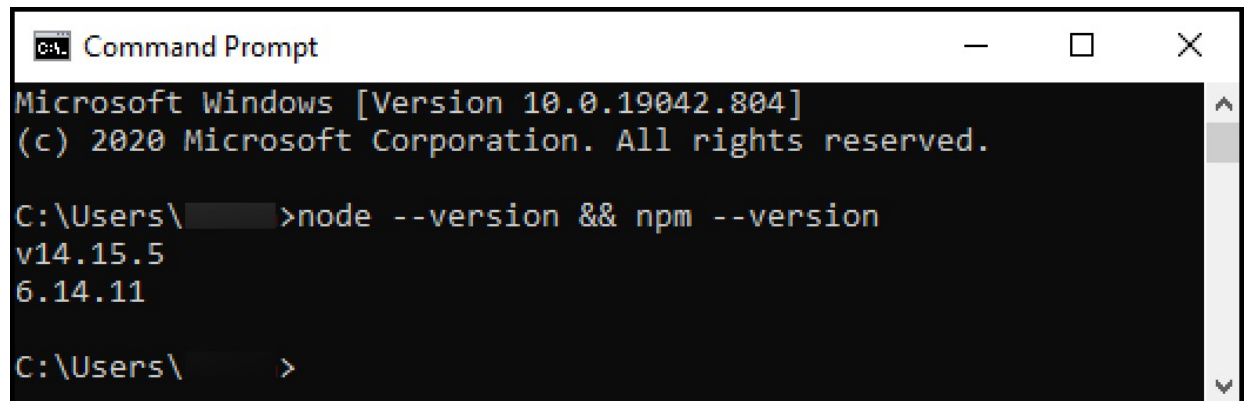
→ Using cmd & macOS terminal:

node --version && npm --version



```
Select Windows PowerShell

PS C:\Users\[redacted]> node --version; npm --version
v14.15.5
6.14.11
PS C:\Users\[redacted]>
```

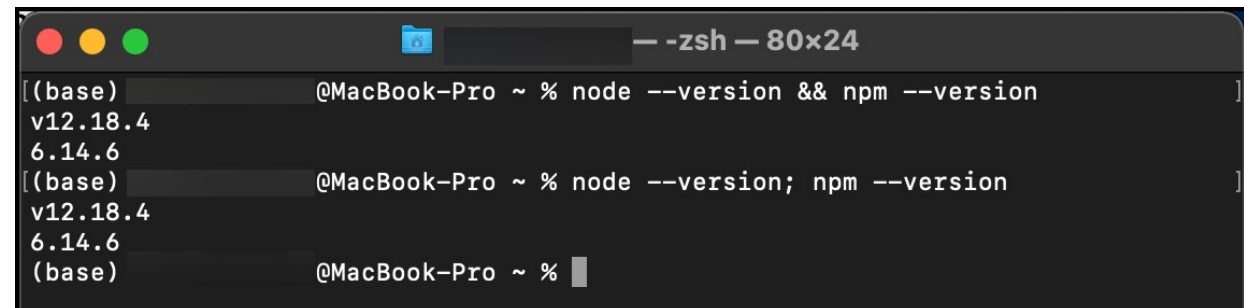


```
Command Prompt

Microsoft Windows [Version 10.0.19042.804]
(c) 2020 Microsoft Corporation. All rights reserved.

C:\Users\[redacted]> node --version && npm --version
v14.15.5
6.14.11

C:\Users\[redacted]>
```



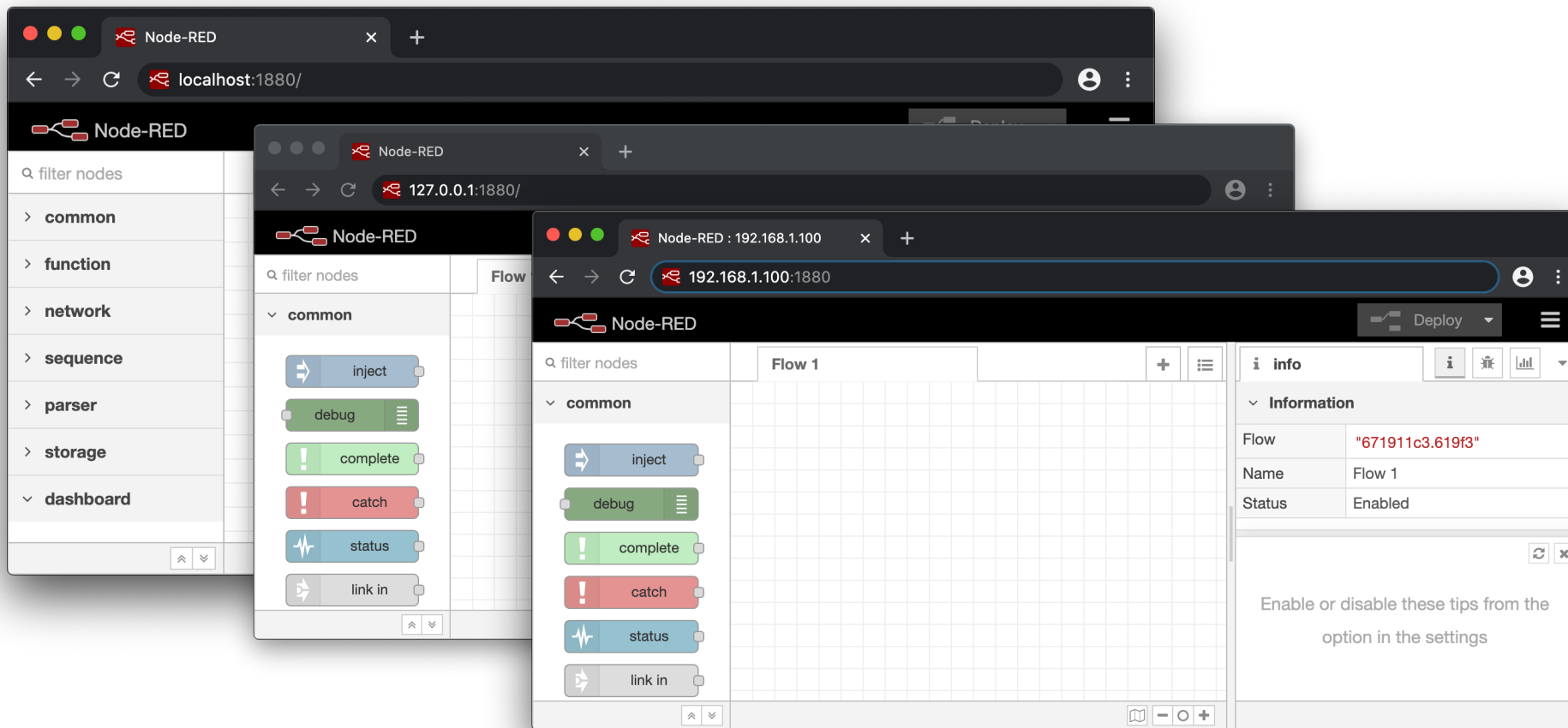
```
- zsh — 80x24

(base) @MacBook-Pro ~ % node --version && npm --version
v12.18.4
6.14.6
(base) @MacBook-Pro ~ % node --version; npm --version
v12.18.4
6.14.6
(base) @MacBook-Pro ~ %
```


Download Node-RED & Install

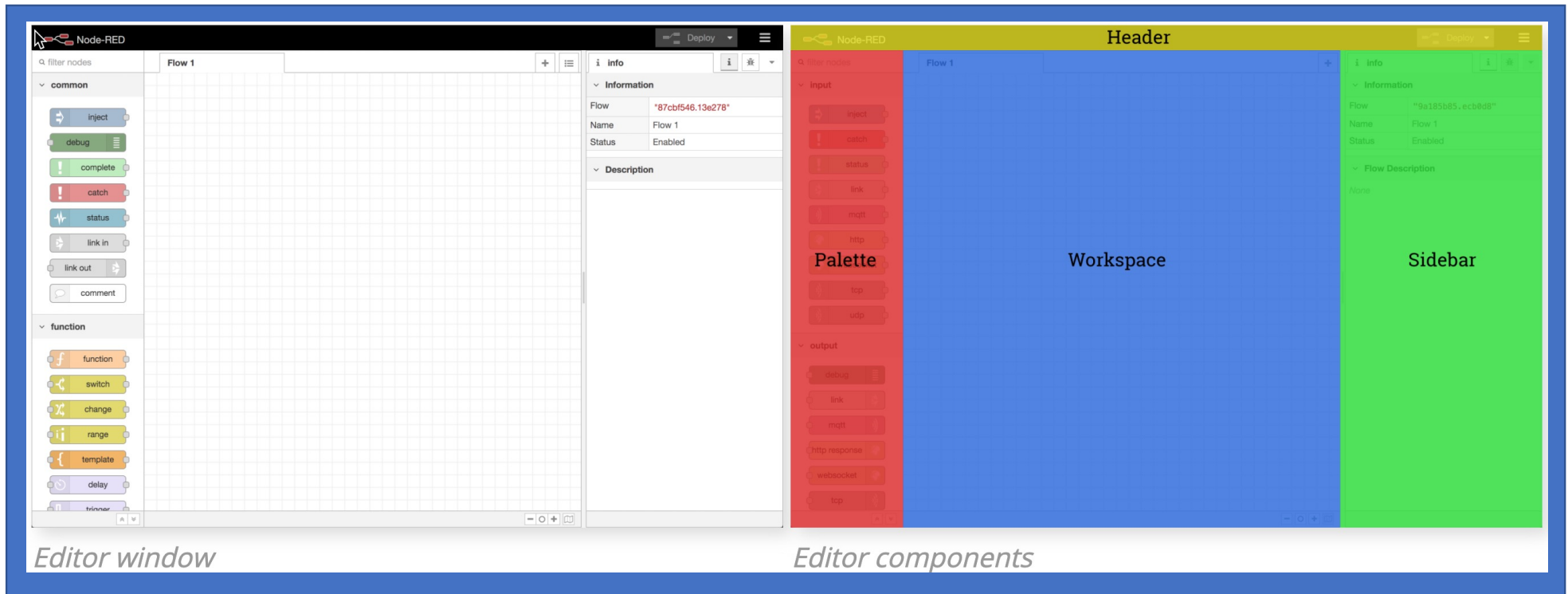
OPENING THE EDITOR

>> You can then access the Node-RED editor by pointing your browser at <http://localhost:1880>.

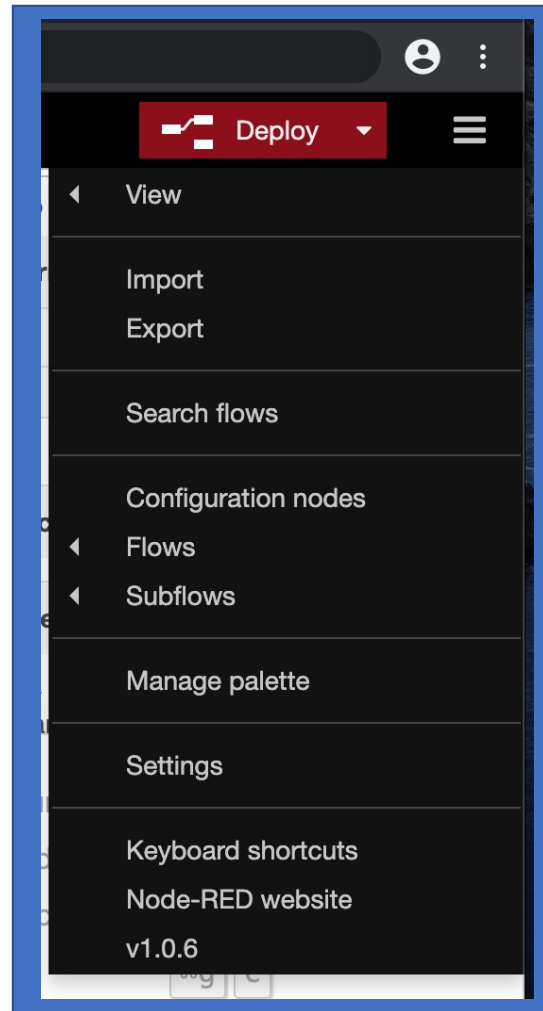


Node-RED: The Editor

The editor components.



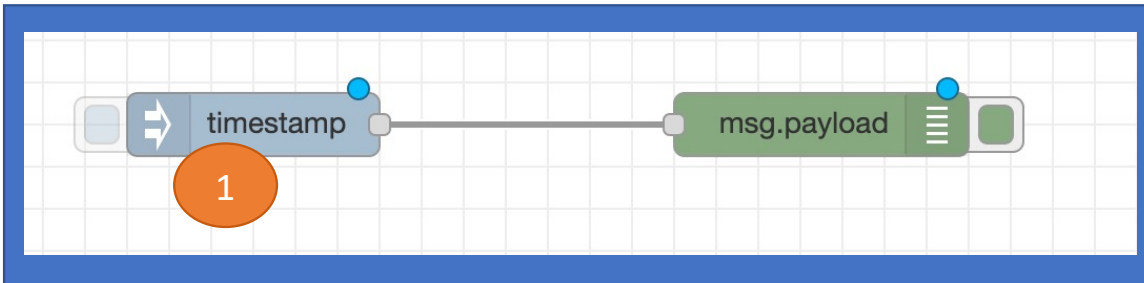
Node-RED: The Menu



EXERCISE 1: Sending Information over NodeRED

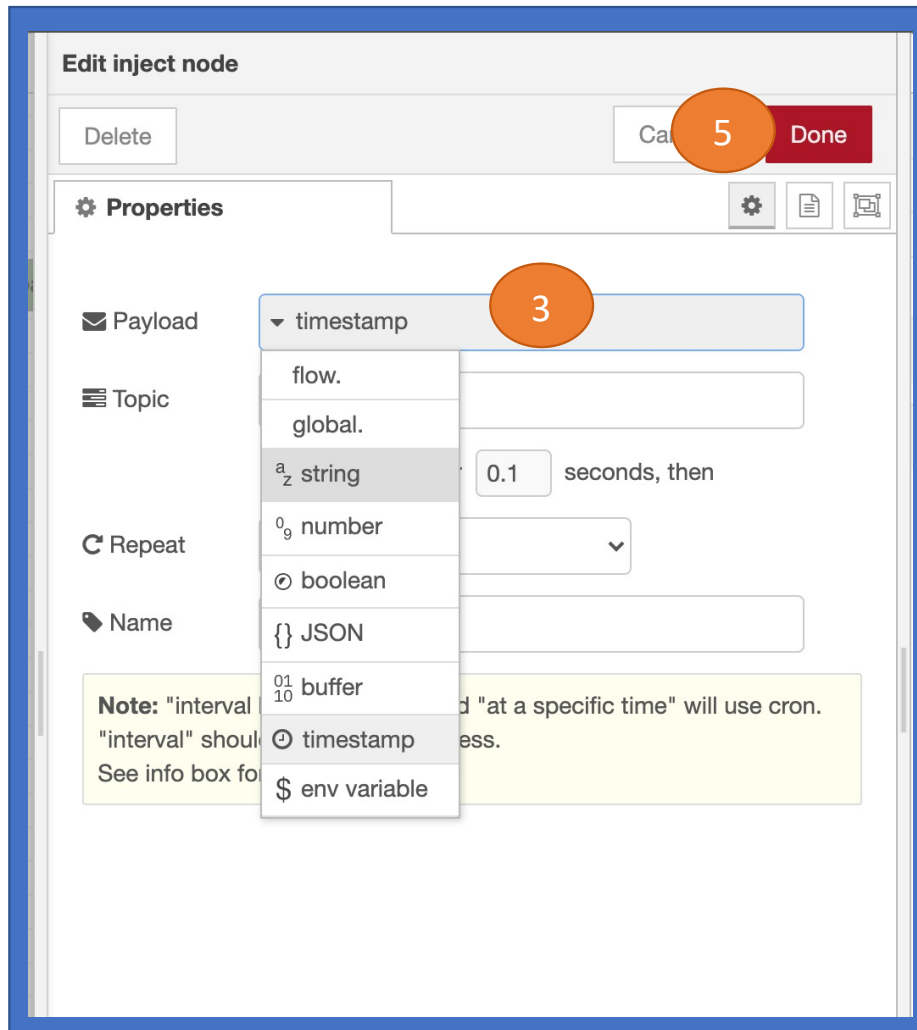
i. Display your name when the inject node is click.

1. Place *Inject node* and *Debug node* into workspace.
2. Double click the *Inject node* to change the properties.



A screenshot of the 'Edit inject node' dialog box in NodeRED. The dialog has a title bar 'Edit inject node' and buttons for 'Delete', 'Cancel', and 'Done'. Below the title bar is a 'Properties' section with a gear icon, a document icon, and a refresh icon. The 'Payload' field is a dropdown menu showing 'timestamp'. The 'Topic' field is an empty text box. There is a checkbox for 'Inject once after' followed by a text box containing '0.1' and the text 'seconds, then'. The 'Repeat' field is a dropdown menu showing 'none'. The 'Name' field is a text box containing 'Name'. At the bottom, there is a yellow note box with the text: 'Note: "interval between times" and "at a specific time" will use cron. "interval" should be 596 hours or less. See info box for details.' An orange circle with the number '2' is placed over the 'Edit inject node' title bar.

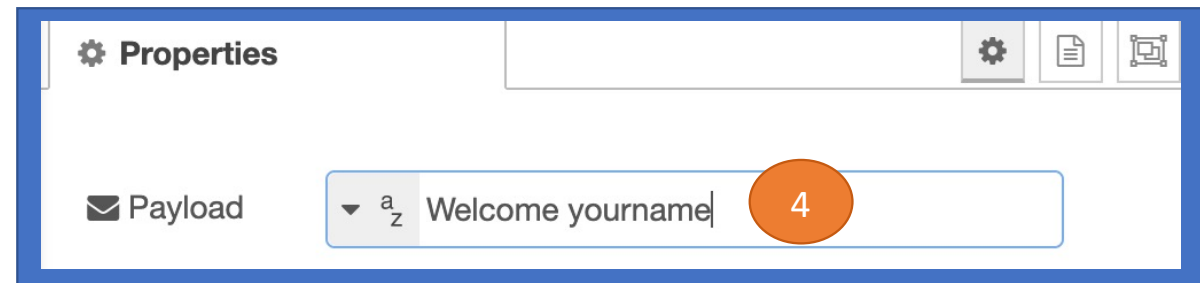
EXERCISE 1: Sending Information over NodeRED



3. Click at the *Payload* dropdown box and choose *string*.

4. Type *Welcome 'yourname'* .

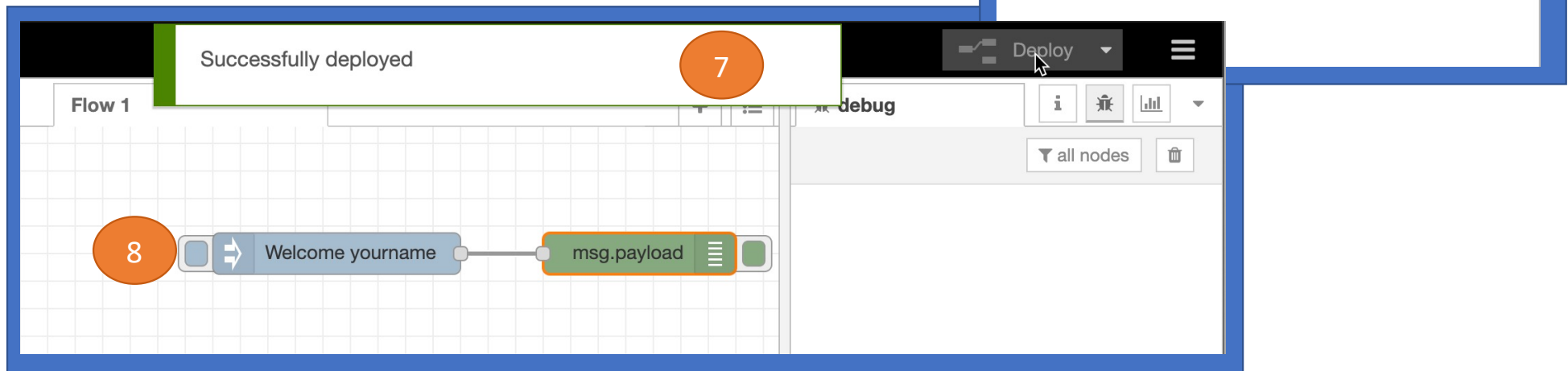
5. Click *Done*.



EXERCISE 1: Sending Information over NodeRED

6. Click *Deploy* to update, save and enable execute. It becomes red when you make any changes.

7. Success message box appears right after click *Deploy*.



8. Click to small blue square to trigger the output.

EXERCISE 1: Sending Information over NodeRED

9. A successful injected message will pop-up immediate you click inject node.
10. Click the bug icon to view the output via debug activity.
11. The output.

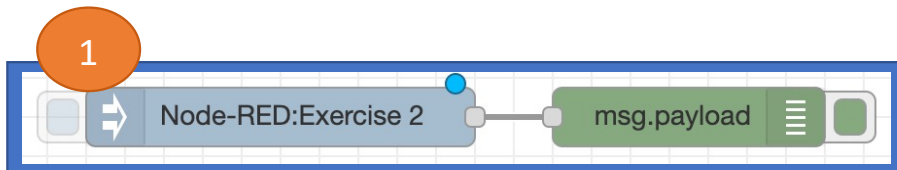
The screenshot displays the NodeRED web interface. At the top, a notification box (labeled 9) states "Successfully injected: Welcome yourname". Below this, the main workspace shows a flow named "Flow 1" with two nodes: a blue "inject" node with the text "Welcome yourname" and a green "msg.payload" node. The right-hand sidebar contains the "debug" tab (labeled 10), which shows the output of the injected message (labeled 11):

```
03/06/2020, 22:01:57 node:  
de416064.1ad6a  
msg.payload : string[16]  
"Welcome yourname"
```

EXERCISE 2: Update Information over NodeRED after certain seconds

ii. Change inject mode

1. Set *Inject node* and *Debug node* as follows.
2. Click *Inject* once after 0.1 seconds, then...
3. Click *Done*.
4. Click *Deploy*.
5. Click *Debug* window.
6. Observe the output.

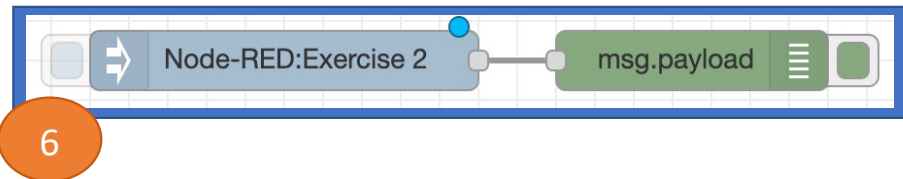


The "Edit inject node" configuration window is shown. It includes fields for Payload (Exercise 2), Topic (Node-RED), and Name (Name). The "Inject once after" checkbox is checked, and the interval is set to 0.1 seconds. The "Repeat" dropdown is set to "none". A note at the bottom states: "Note: 'interval between times' and 'at a specific time' will use cron. 'interval' should be 596 hours or less. See info box for details." An orange circle with the number "2" is positioned over the "Inject once after" checkbox.

EXERCISE 3: Update Information over NodeRED with Every 1 second

iii. Change inject mode

1. Set *Inject node* and *Debug node* as follows.
2. Set *Repeat* > *interval* > *every 1* > *seconds*.
3. Click *Done*.
4. Click *Deploy*.
5. Click *Debug* window.
6. Go to workspace & click Inject button.
7. Observe the output.

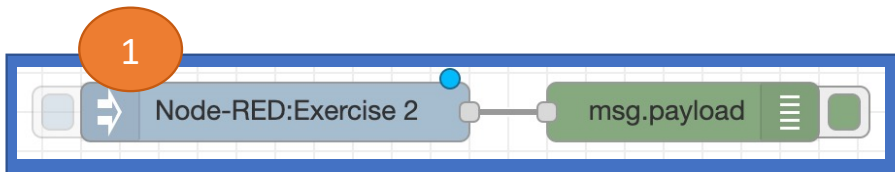


The 'Edit inject node' dialog box is shown. It has a 'Delete' button, a 'Cancel' button, and a 'Done' button. The 'Properties' tab is selected. The 'Payload' field is set to 'Exercise 2'. The 'Topic' field is set to 'Node-RED'. The 'Repeat' section is expanded, showing 'interval' selected from a dropdown menu. The 'every' field is set to '1' and the unit is 'seconds'. A blue circle with the number '2' is overlaid on the 'interval' dropdown menu. A note at the bottom states: 'Note: "interval between times" and "at a specific time" will use cron. "interval" should be 596 hours or less. See info box for details.'

EXERCISE 4: Update Information over NodeRED with Every 1 second

iv. Enabled vs Disabled mode

1. Set *Inject node* and *Debug node* as follows.
2. Click *Enabled* radio button.
3. Report the results.



The 'Edit inject node' configuration dialog is shown. It has a title bar 'Edit inject node' and buttons for 'Delete', 'Cancel', and 'Done'. The 'Properties' tab is active. The configuration includes:

- Payload:** A dropdown menu showing 'a_z Exercise 2'.
- Topic:** A text input field containing 'Node-RED'.
- Inject once after:** A checkbox that is unchecked, followed by a text input '0.1' and the word 'seconds, then'.
- Repeat:** A dropdown menu set to 'interval'.
- every:** A text input '1' followed by a dropdown menu set to 'seconds'.
- Name:** A text input field containing 'Name'.

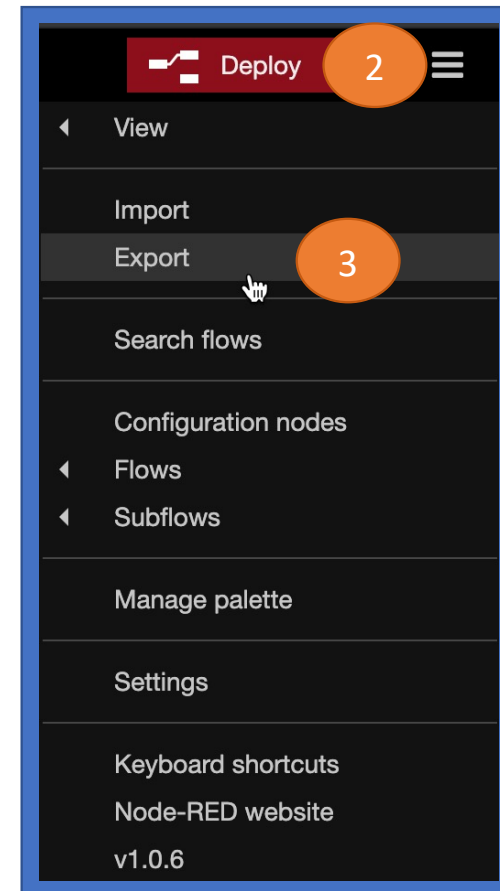
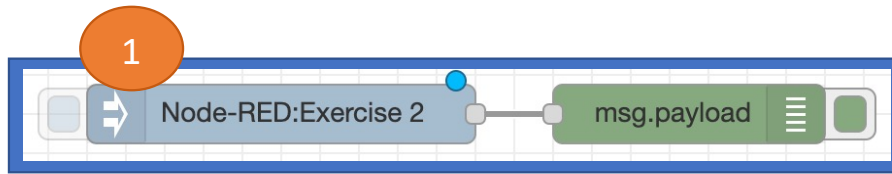
A yellow note box at the bottom states: "Note: 'interval between times' and 'at a specific time' will use cron. 'interval' should be 596 hours or less. See info box for details."

At the bottom left, there is a radio button labeled 'Enabled' which is selected. An orange circle with the number '2' is positioned next to it.

EXERCISE 5: Exporting Your Works

v. Exporting your works

1. Set *Inject node* and *Debug node* as follows.
2. Click *Menu*.
3. Click *Export*.



EXERCISE 5: Exporting Your Works

v. Exporting your works

4. Click *Copy to clipboard*.
Click *Download* if you wish to download the nodes and it will be saved into your laptop/hard disk with the name of *flows.json*.

Export nodes

Export selected nodes current flow all flows

Clipboard

Library

```
[{"id":"7fdeaeda.d2619","type":"tab","label":"Flow 1","disabled":false,"info":""}, {"id":"a123353c.973608","type":"inject","z":"7fdeaeda.d2619","name":"","topic":"Node-RED","payload":"Exercise 2","payloadType":"str","repeat":"1","crontab":"","once":false,"onceDelay":0.1,"x":240,"y":100,"wires":[["de416064.1ad6a"]]}, {"id":"de416064.1ad6a","type":"debug","z":"7fdeaeda.d2619","name":"","active":true,"tosidebar":true,"console":false,"tostatus":false,"complete":"payload","targetType":"msg","x":450,"y":100,"wires":[]}]
```

compact formatted

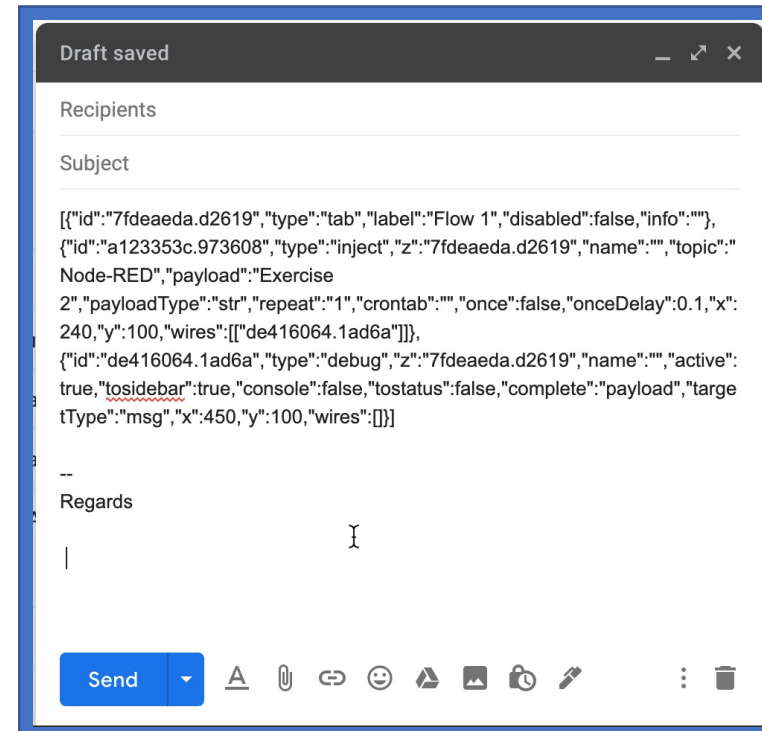
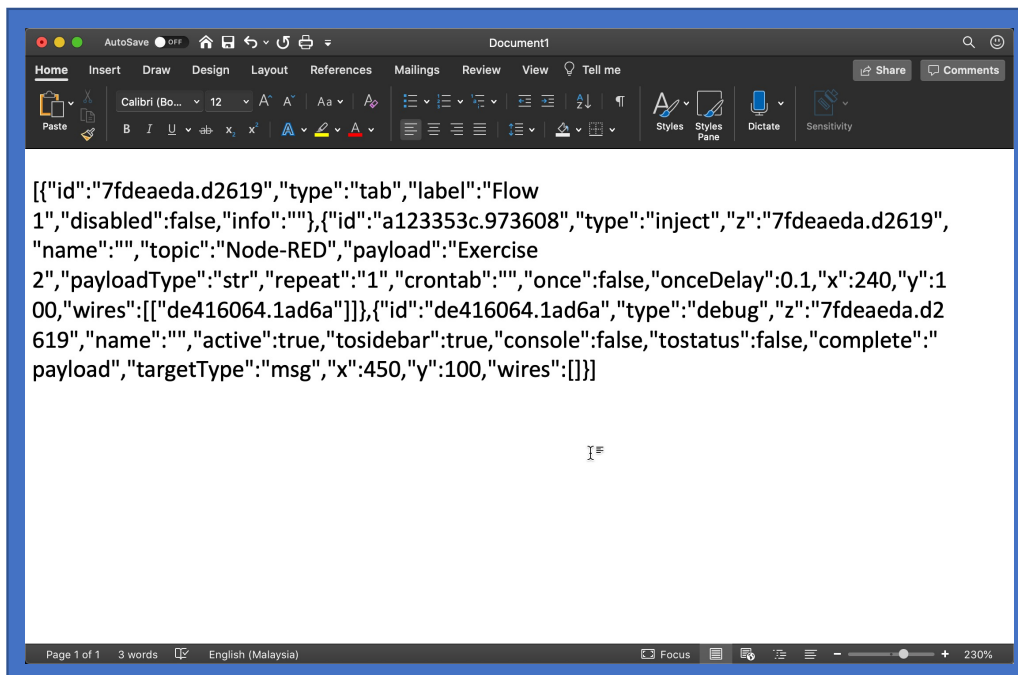
4

Cancel Download Copy to clipboard

EXERCISE 5: Exporting Your Works

v. Exporting your works

6. Open any text editor, email and paste work.



QUESTIONS

END